Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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Course Title: Digital Logic Design Laboratory

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Experiment No: 10

Experiment Title: Design and implementation of a 2 bit asynchronous counter.

Submitted To:

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Experiment No: 10

Experiment Name:

Design and implementation of a 2 bit asynchronous counter using JK flipflops.

Objective:

To design and implement a 2-bit asynchronous counter and verify the sequence (00→01→10→11)

Required Components:

Breadboard, Digital trainer board, IC 7476, connecting wires, clock signal.

Theory:

An asynchronous counter is a sequential circuit where the flip flops are not clocked simultaneously.

For a 2 bit counter,

FF0 (LSB) : Toggles on each clock pulse.

FF1 (MSB) : Toggles when Q_0 from 1→0

Sequence: 00→01→10→11→00. . . .

MOD : 2^2 (4 states)

Circuit Diagram:

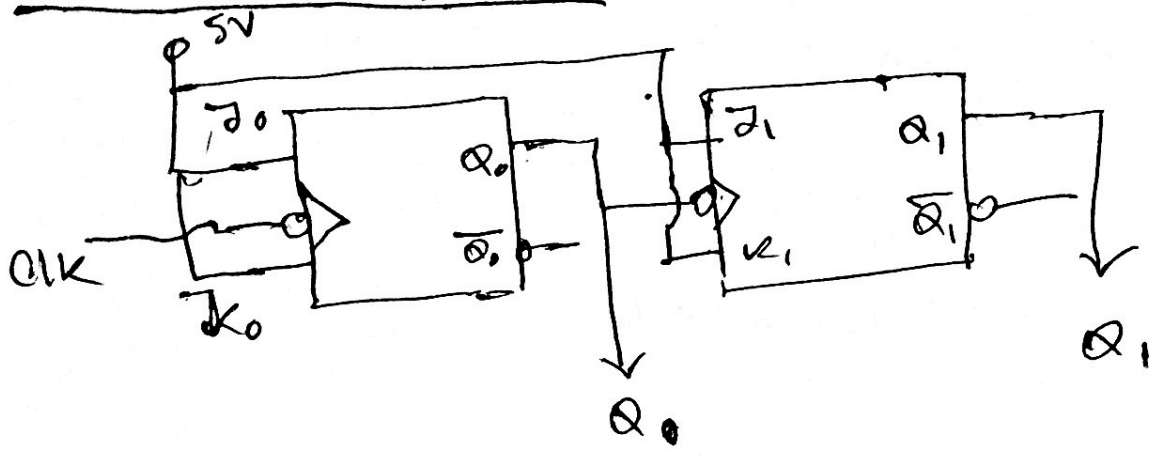


Fig: 2 bit Asynchronous flipflop

Procedure

1. Set up the circuit on the board.
2. connect the flipflop IC to the inputs and output
3. Connect the clock pulse and check truth Table

Truth Table:

clk	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

Result:

- Maintained the sequence 00→01→10→11→00.
- LSB toggled in each clock pulse.
- MSB toggled when LSB = 1.

Discussion:

The asynchronous counter worked as expected, demonstrating binary counting.